

Chinmay Kapoor

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Forward Deployed GenAI Engineer | Agentic AI Systems | Customer-Facing Production AI

SUMMARY

AI/ML engineer with 3+ years building applied AI systems and 2+ years delivering LLM, RAG, and agentic workflows for customer-facing environments. Experienced taking prototypes into customer-ready deployments by connecting models to APIs, private data, local/cloud runtimes, governed tool execution, session/state management, evaluation, and traceable debugging. Strong in Python, FastAPI, JavaScript/TypeScript, vector search, LLM routing, technical demos, and cross-functional delivery with software, data, sales, hardware, and customer teams.

EXPERIENCE

Field Application Engineer - AI Specialist, MIMIK (Vancouver, BC) Oct 2025 - Present

- Built **MimOE Studio** to make agentic AI demos deployable across customer environments, routing cloud/local models through FastAPI/SSE with session state, provider fallback, and web/mobile controls for field and internal workflows.
- Reduced deployment risk and improved agent safety for real tool use by converting model decisions into typed file/search/shell actions guarded by workspace boundaries, approval checks, risk policy, and audit logs instead of unrestricted model access.
- Standardized multi-agent and external-tool integration through MCP-style tool exposure and ADK-compatible runtime paths, giving agents a governed action contract for file, search, shell, and API workflows across local, cloud, and customer-controlled environments.
- Preserved session context across model-agnostic routes including Ollama, OpenAI-compatible APIs, CLI-backed models, local runtimes, cloud endpoints, and customer-controlled environments.
- **Blackbox View:** Built evaluation and observability views for API calls, session state, model/tool rounds, context injection, latency points, and agent failure paths, making system behavior explainable during demos and deployment reviews.
- Reduced setup friction across Linux, ARM, x86, and edge machines by building environment-aware runtime services for local model detection, GGUF scan/download, llama.cpp/llama-server setup, and runtime controls.
- Improved agent and developer grounding by building tree-sitter-based context indexing with SQLite-backed code-graph views for APIs, functions, files, and dependencies before code or configuration changes.
- Improved incident-handling accuracy by 30% by testing BLE/location-triggered response logic against missed-location, false-zone, delayed-response, and cascading-failure cases before events activated local AI workflows.
- Turned backend implementation details into customer-ready enablement by creating deployment guides, user demos, architecture walkthroughs, and product videos used on MIMIK social channels and in online/event demos.

Data Scientist - Lead / AI Consultant, Swissvault Global (Vancouver, BC) Jan 2024 - Oct 2025

- **COMPIO:** Helped customers choose cloud, on-prem, or hybrid AI architectures by building a benchmarking tool that replayed real workflows from API/web ingestion and object storage through RAG, retrieval, and inference.
- **SVSI:** Enabled secure enterprise data access by building a local-first RAG interface for VaultFS, letting business users query sensitive files through local LLMs with optional cloud API offload for non-sensitive tasks.
- Bridged legacy/private storage with GenAI workflows by connecting VaultFS, AWS S3, Azure/Fabric, vector databases, local models, and OpenAI/Gemini-style APIs into one governed retrieval and inference path for customer-controlled data environments.
- Reduced edge inference latency and lowered OPEX by 21% by evaluating retrieval quality, quantization tradeoffs, and hardware/runtime fit across CPU/GPU/NPU setups, then applying INT4/INT8 and offload decisions.
- Accelerated POC-to-MVP delivery by 45% by translating client requirements into scoped features with software, hardware, sales, and data engineering teams, supported by whitepapers, FAQs, performance reports, and technical documentation.
- Built Imagilect computer-vision retrieval features using YOLOv5, DeepSORT, OCR, and ResNet50, linking model outputs to structured storage so users could search and analyze video-derived records.

Research Assistant (ML Team Lead), SOLIDS Lab, UVIC (Victoria, BC) Sep 2021 - Aug 2024

- Developed **FUSTT**, a spatio-temporal transformer for long-term multivariate forecasting on real-world sensor datasets for an NRC-sponsored digital twin initiative, improving forecasting accuracy by 16% and compute time by 19%.
- Designed and validated forecasting pipelines with Transformers, DLinear, LSTM-based baselines, ablation studies, temporal feature engineering, missing-data handling, and MSE/MAE model evaluation.
- Published 3 peer-reviewed works across IEEE and Springer venues, filed or received 3 grants, and presented technical findings at CSCE 2022 and UVic/OCEAN CleanTech events.

SKILLS

Agentic AI & LLM Systems : multi-agent workflows, Model Context Protocol (MCP)-style tool integration, ADK-compatible runtime work, tool/function calling, RAG, ReAct-style tool loops, state/session management, LangChain, LangGraph, PydanticAI, Qdrant, ChromaDB, FAISS, Ollama, Hugging Face, llama.cpp

Production Engineering : Python, JavaScript, TypeScript, FastAPI, Starlette, REST APIs, SSE streaming, Docker, Kubernetes/GKE, Cloud Run concepts, Git/GitHub Actions, CI/CD, Linux, Bash, SQLite, Pydantic, JSONL tracing, audit logs, monitoring, deployment debugging

Evaluation & Data : latency/retrieval evaluation, model benchmarking, tokens/usage tracking, prompt and tool-call debugging, PyTorch, TensorFlow, ONNX, scikit-learn, pandas, NumPy, PySpark, SQL, BigQuery, Gemini API integration patterns, Vertex AI concepts, Microsoft Fabric, AWS S3

Customer-Facing Delivery : forward-deployed demos, architecture walkthroughs, production-readiness reviews, deployment guides, whitepapers, FAQs, POC scoping, field feedback, product/engineering handoff, sales/FAE collaboration

EDUCATION

M.Eng. ECE – *University of Victoria* (2021–2024)

B.Eng. CSE – *Amity University* (2017–2021)

PUBLICATIONS

- **FUSTT**[†] – Spatio-temporal transformer for multivariate long-term forecasting of marine stressors, sponsored by NRC and published through IEEE/NRC.
- **Concrete Characteristics Evaluation**[†] – Machine learning prediction of concrete properties for sustainability-focused material analysis and carbon-emission reduction.
- **Human Activity Recognition**[†] – Evaluated human activity detection using wearable sensor data and LSTM-based sequence modeling.

SELECTED PROJECTS

- **COMPIO**[†] – FastAPI/RAG market-intelligence platform using API/web ingestion, vector search, S3 storage, and structured report generation.
- **Fraud Detection ML**[†] – Classification pipeline using Logistic Regression and Random Forest with class-imbalance handling, confusion-matrix analysis, precision, recall, and F1 evaluation.
- **Stock Forecasting**[†] – Time-series forecasting workflow comparing regression, LSTM, and XGBoost baselines with feature engineering, prediction visualization, and app-based outputs.